

# PRACTICE QUIZ

40 multiple choice · 15 true/false · 3 matching sets — his stated format. Closed book, 6:35pm.

**Work it closed-book and timed.** Write your answers on paper — the point is retrieval, not recognition. The answer key explains the reasoning for every item, so mark yourself honestly and go read the explanation for anything you missed or guessed.

## Part A · Multiple Choice

1. Blood is classified as a connective tissue because:

- A) It has a matrix with cells suspended in it
- B) It is produced inside bone
- C) It contains collagen fibres
- D) It connects organs to one another

2. The normal pH range of blood is:

- A) 7.25 – 7.35
- B) 7.35 – 7.45
- C) 7.45 – 7.55
- D) 7.00 – 7.20

3. Blood viscosity is determined primarily by:

- A) Plasma protein concentration
- B) White blood cell count
- C) The number of red blood cells
- D) Water content of plasma

4. After centrifugation, plasma represents approximately what fraction of blood volume?

- A) 45%
- B) 90%
- C) 70%
- D) 55%

5. The dominant protein in plasma is:

- A) Albumin
- B) Thrombin
- C) Fibrinogen
- D) Gamma globulin

6. The primary function of albumin is to:

- A) Transport oxygen
- B) Maintain osmotic pressure
- C) Act as an antibody
- D) Form the clot mesh

7. Most plasma proteins are manufactured by the:

- A) Red bone marrow
- B) Kidneys
- C) Liver
- D) Spleen

8. The swollen abdomen seen in Kwashiorkor is caused by:

- A) Enlarged liver
- B) Intestinal parasites
- C) Excess fat deposition
- D) Fluid leaving the circulation into the tissue

9. Mature erythrocytes lack all of the following EXCEPT:

- A) Hemoglobin
- B) A nucleus
- C) Mitochondria
- D) Ribosomes

10. The biconcave shape of the RBC is advantageous because:

- A) It makes the cell heavier than plasma
- B) No interior point lies far from the membrane, speeding gas exchange
- C) It protects the nucleus
- D) It allows the cell to store more iron

11. Red blood cells generate ATP anaerobically. The functional consequence is:

- A) They can survive without glucose
- B) They produce ATP more efficiently than other cells
- C) They do not consume the oxygen they carry
- D) They divide more rapidly

12. Oxygen binds to which part of the hemoglobin molecule?

- A) The globin protein chains
- B) The ribosomes
- C) The cell membrane
- D) The iron atom at the centre of each heme

13. One red blood cell can carry approximately how many oxygen molecules?

- A) 1 billion
- B) 250 million
- C) 4
- D) 250 thousand

14. The average lifespan of an erythrocyte is:

- A) 30–40 days
- B) 100–120 days
- C) 1 year
- D) 60–80 days

15. Aged and damaged red blood cells are removed primarily by the:

- A) Kidneys
- B) Liver
- C) Spleen
- D) Red bone marrow

16. Erythropoietin is secreted by the \_\_\_\_\_ in response to \_\_\_\_\_.

- A) Red bone marrow ... blood loss
- B) Kidneys ... hyperoxia
- C) Liver ... hypoxia
- D) Kidneys ... hypoxia

17. A reticulocyte is best described as:

- A) An immature RBC that has ejected its nucleus but retains ribosomes
- B) A white blood cell precursor
- C) An RBC that has lost its hemoglobin
- D) A fragment of a megakaryocyte

18. A patient's reticulocyte count is 10%. This most likely indicates:

- A) A primary bone marrow disease
- B) The bone marrow is compensating for a problem elsewhere
- C) Normal healthy variation
- D) Vitamin B12 toxicity

19. Hematopoiesis occurs in:
- A) Yellow bone marrow
  - B) The liver in adults
  - C) Red bone marrow
  - D) The spleen
20. Which is NOT a typical site of adult hematopoiesis?
- A) Proximal femur
  - B) Pelvic girdle
  - C) Vertebrae
  - D) Distal tibia
21. Which nutrient sits at the centre of the heme group?
- A) Iron
  - B) Folate
  - C) Calcium
  - D) Vitamin B12
22. The most abundant white blood cell is the:
- A) Monocyte
  - B) Neutrophil
  - C) Lymphocyte
  - D) Basophil
23. An elevated eosinophil count most strongly suggests:
- A) Bone marrow failure
  - B) Viral infection
  - C) Parasitic infection or allergy
  - D) Bacterial infection
24. Which white blood cell develops into a macrophage in the tissues?
- A) Neutrophil
  - B) Eosinophil
  - C) Basophil
  - D) Monocyte
25. Histamine is released primarily by:
- A) Basophils
  - B) Erythrocytes
  - C) Monocytes
  - D) Neutrophils
26. The ability of a white blood cell to squeeze through a vessel wall into tissue is called:
- A) Phagocytosis
  - B) Diapedesis
  - C) Hemostasis
  - D) Chemotaxis
27. Leukopenia is most commonly induced by:
- A) High altitude
  - B) Stress
  - C) Chemotherapy
  - D) Bacterial infection
28. Platelets are formed from:
- A) Reticulocytes
  - B) Monocytes
  - C) Lymphoblasts
  - D) Megakaryocytes

29. The hormone that regulates platelet production is:
- A) Thrombopoietin
  - B) Erythropoietin
  - C) Nitric oxide
  - D) Thyroxine
30. Platelets are normally kept inactive in circulation by:
- A) Heparin secreted by basophils
  - B) Nitric oxide secreted by endothelial cells
  - C) Albumin
  - D) Low blood calcium
31. Place the steps of hemostasis in the correct order:
- A) Vascular spasm → fibrin mesh → platelet plug
  - B) Fibrin mesh → vascular spasm → platelet plug
  - C) Vascular spasm → platelet plug → fibrin mesh
  - D) Platelet plug → vascular spasm → fibrin mesh
32. Which conversion produces the insoluble threads that seal a wound?
- A) Albumin → globulin
  - B) Prothrombin → thrombin
  - C) Heme → bilirubin
  - D) Fibrinogen → fibrin
33. A clot that forms and remains in an unbroken vessel is called a(n):
- A) Thrombus
  - B) Embolus
  - C) Infarct
  - D) Hematoma
34. A patient with a known DVT is at greatest risk of pulmonary embolism when they:
- A) Sleep
  - B) Begin walking
  - C) Remain seated
  - D) Elevate the legs
35. Low-dose aspirin prevents clots by:
- A) Inhibiting vitamin K
  - B) Dissolving existing clots
  - C) Blocking platelet aggregation
  - D) Destroying platelets
36. Warfarin (Coumadin) works by:
- A) Inhibiting thrombin directly
  - B) Blocking platelet aggregation
  - C) Increasing nitric oxide
  - D) Interfering with vitamin K
37. Newborns receive a vitamin K injection at birth because:
- A) Their gut has not yet been colonised by the bacteria that produce it
  - B) It prevents jaundice
  - C) Breast milk lacks vitamin K entirely
  - D) Their liver cannot process vitamin K
38. Severe liver disease causes a bleeding tendency because the liver:
- A) Filters old red blood cells
  - B) Manufactures clotting factors, which are proteins
  - C) Stores platelets
  - D) Produces erythropoietin

39. Secondary polycythemia at high altitude is best described as:

- A) An autoimmune disease
- B) A form of anemia
- C) A compensatory physiological response
- D) A bone marrow cancer

40. After withdrawing an acupuncture needle that produces a drop of blood, you should:

- A) Leave the point uncovered
- B) Press and gently massage the point
- C) Apply firm massage for 30 seconds
- D) Press and hold with a cotton ball

## Part B · True / False

Several of these are built from misconceptions the professor corrected in class. If one feels obviously true, look twice.

- 1. Blood thinners work by reducing the number of red blood cells. T / F
- 2. Aspirin reduces the ability of platelets to aggregate without destroying them. T / F
- 3. Atherosclerosis is caused primarily by plaque buildup rather than by blood viscosity. T / F
- 4. An elevated reticulocyte count is a compensatory response rather than a disease. T / F
- 5. Thyroid hormone directly regulates erythropoiesis. T / F
- 6. Neutrophils, lymphocytes, monocytes, eosinophils and basophils are all granulocytes. T / F
- 7. Natural killer cells contain granules that are simply not visible under light microscopy. T / F
- 8. A white blood cell count alone can be used to indicate sepsis. T / F
- 9. A stressful day can raise a person's white blood cell count. T / F
- 10. Plasma proteins serve as an important fuel source for blood cells. T / F
- 11. Increasing nitric oxide as much as possible improves platelet function. T / F
- 12. The red blood cell makes ATP anaerobically and does not consume the oxygen it carries. T / F
- 13. A cell committed to the myeloid pathway can later become a lymphocyte. T / F
- 14. Lymph, like blood, is classified as a connective tissue. T / F
- 15. Platelets are true cells with a nucleus and full organelles. T / F
- 16. Vascular spasm alone is sufficient to stop bleeding from a damaged vessel. T / F

## Part C · Matching

1. Match the hormone to its source and target.

| Term           | Your answer |
|----------------|-------------|
| Erythropoietin | _____       |
| Thrombopoietin | _____       |

- A. Kidney → red bone marrow; triggered by hypoxia
- B. Liver/kidney → megakaryocytes; regulates platelet output

2. Match each drug to its mechanism.

| Term     | Your answer |
|----------|-------------|
| Aspirin  | _____       |
| Heparin  | _____       |
| Warfarin | _____       |

- A. Blocks platelet aggregation
- B. Inhibits thrombin; injectable, used in hospital

C. Interferes with vitamin K; oral, used after discharge

3. Match each anemia to its cause family.

| Term                   | Your answer |
|------------------------|-------------|
| Iron deficiency anemia | _____       |
| Aplastic anemia        | _____       |
| Sickle cell anemia     | _____       |
| Heavy menstruation     | _____       |

- A. Blood loss
- B. Production fails
- C. Production fails — marrow stops making all lines
- D. Destruction / defect

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**Scoring guide.** Under 80% — read the guide's ★ boxes again before bed. 80–90% — you're in good shape; drill the △ traps, since that's where the true/false marks live. Above 90% — sleep.